UC San Diego
EXTENDED STUDIES

Course Syllabus

Course Title:	Embedded Systems Hardware Design
Instructor Name:	Saleem Yamani
Course Number:	ECE-40292
Section ID:	194177
Quarter:	Winter 2026
Course Dates:	1/6/26-3/7/26

Welcome to Embedded Systems Hardware Design!

I am excited to work with you in this course and to help you build a strong foundation in designing real-world embedded systems. This is a hands-on, industry-focused course, and I encourage you to ask questions, explore the material, and actively engage in the labs and assignments. I look forward to learning and working together with you throughout the course.

Communication Policy

I want to hear from you! It's not unusual to encounter difficulties with course content or assignments. If that happens to you, please reach out to me as soon as possible so that we can begin to solve the problem. The earlier you communicate with me about a potential issue, the more likely we are to find a good solution. I will do my best to respond to messages as soon as possible and within 48 hours, unless otherwise stated.
Click on 'Help' in the universal navigation menu anywhere in Canvas to visit the *Canvas Student Orientation* course to see guidelines for appropriate digital communication etiquette between students and instructors.

Course Description

Developing an embedded system requires knowledge in both software and hardware design. The embedded systems engineering industry is showing an increasing demand for software engineers with a background in hardware design.

Embedded systems can be found in many industries, including consumer electronics, automotive, aviation, transportation, agriculture, medical devices, and scientific research. It is also widely used in industrial automation, including robotics, motion control systems, and Supervisory Control and Data Acquisition (SCADA) systems.

This course is intended for technical professionals looking to acquire solid foundational knowledge necessary to design embedded systems. Students will learn definition and main components (modules) of embedded systems, hardware design aspects, Commercial off the Shelf (COTS) embedded

Course Prerequisites

ECE-40153 Introduction to Embedded Systems or equivalent knowledge and experience. Knowledge of electronic components and modules and the ability to read and understand electrical schematics required for this course. Working knowledge of KiCAD and LTSpice will be very helpful to students in this course but not required.

Student Learning Outcomes

Student Learning Outcomes are specific and measurable goals for student learning. Students demonstrate their mastery of these outcomes through course assessments.
By the end of this course, students will be able to:

- Construct a block diagram of the required embedded system
- Design proper interfaces between the modules of an embedded system
- Design the required sensors and actuators
- Building and testing an embedded system

If you are unclear about the outcomes for this course or their application, or if you would like more information about them, please reach out to me.

Course Materials

Required Materials. The following required materials are essential to your learning and will be actively used during this course.
|

- Purchase STMicroelectronics **Discovery Kit B-L475E-IOT01A1** from:
  http://www.st.com/en/evaluation-tools/b-l475e-iot01a.html
  or buy it from distributors such as Mouser, Digikey, etc

- Wolf, M. (2022). **Computers as Components:** principles of embedded computing system design. Waltham, MA: Morgan Kaufmann/Elsevier. (**Print ISBN-13:** 978-0323851282, **eBook ISBN-13:** 978-0323851299)

Course Schedule

Below, please find the schedule for course topics and assignments. These dates are an estimate to help you with your planning. Please note that this schedule is subject to change during the course.

Session	Topics & Reading	Assignments (see Canvas for due dates)
1	**Embedded System Architecture & System-Level Design** **Key Topics:** Introduction to the course syllabus	**Purchase STM32 board and download documentation** **Written Assignment:** Assignmnet#1 **Labs:** LAB#1

	• Post self-introduction message to discussion forum • Introduction to Principles of Embedded Systems **Reading:** Read section 1.2 from textbook Reading additional material provided in the module	
2	**Embedded System Design Process & COTS Modules** **Key Topics:** • Describe the importance of Embedded System Abstraction • Identify the key components required in an Embedded System Design • Evaluating Commercial-Off-The-Shelf modules • BOM generation and hardware selection strategies **Reading:** Read section 1.3 from textbook Reading material provided in the module	**Written Assignment:** Quiz#1 **Labs:** Lab#2
3	**Selecting Microcontrollers** **Key Topics:** • Computer Architecture Basics • Compare the different type of processor architectures • ARM Cortex-M architecture **Reading:** Read section 2.2, 2.3, 3.2, 3.3 from textbook Reading material provided in the module	**Written Assignment:** Quiz#2 **Labs:** None
4	**Caches and Memory Management Units (MMU)** **Key Topics:** • Identify different forms of Cache algorithms	**Written Assignment:** Quiz#3 **Labs:** Lab#3

	• Discuss the advantages of having a MMU • Types of embedded memory **Reading:** Read section 3.5, 3.6 from textbook Reading material provided in the module	
5	**Bus based CPU Systems and Sensors** **Key Topics:** • Design and develop using CPU based systems • Sensor categories and characteristics • Reading sensor datasheets **Reading:** Read section 4.2,4.3,4.4, and 4.8 from textbook Reading material provided in the module	**Written Assignment:** Quiz#4 **Labs:** Lab#4
6	**Internet Of Things Systems and Automotive** **Key Topics:** • Internal buses: AHB, APB, DMA • External interfaces: I²C, SPI, UART, USB, CAN • Wireless interface overview (BLE, Wi-Fi, Sub-GHz) • Interface selection based on bandwidth, latency, power **Reading:** Read section 8.2,8.3,8.4,9.2,9.3,9.4 from textbook Reading material provided in the module	**Written Assignment:** Quiz#5 **Labs:** Lab#5
7	**PCB Design Foundations & Hardware Integration** **Key Topics:** • PCB stackups, routing rules, decoupling • Power integrity and grounding techniques	**Written Assignment:** Quiz#6 **Labs:** Lab#6

	• High-speed design considerations • Understanding eval board PCBs using STM32 kit as reference **Reading:** Reading material provided in the module	
8	**System Design Techniques** **Key Topics:** • Summarize how proper design methodology can speed up time to market and improve quality & cost • Identify various Design Methodologies and understand which suits best for your needs • Explain the benefits of having good Requirements and Specifications • Explain why Quality needs to be built into the product **Reading:** Reading material provided in the module	**Written Assignment:** Quiz#7 **Labs:** None
9	**Capstone: Build & Test a Complete Embedded System** **Key Topics:** • System integration strategies • Power and performance optimization • Preparing engineering documentation • Common hardware/firmware integration pitfalls **Reading:** Reading material provided in the module	**Final Project:** Information provided in the module

Grading Policies

Letter grades are based on the UC San Diego Extended Studies Grading Scale. Your final course grade is based on the percentage of points you have earned.

Passing Grades		Non-Passing Grades	
A+	100%	**D**	60-69%
A	93-99%	**F**	59% and below
A-	90-92%		
B+	87-89%		
B	83-86%		
B-	80-82%		
C+	77-79%		
C	73-76%		
C-	70-72%		

For this class to count towards a certificate program, you must select either *Letter Grade* or *Pass/No Pass* as your grading option. If you select Not for Credit (NFC) as your grading option, your completion of the class will not count towards the certificate. If you'd like to change your grading option for this course, you may do so through your MyExtension portal before the final class meeting (or for online courses, by 11:59 p.m. on the day before the published end date the course) or before final grades are posted, whichever comes first.

Academic Integrity

All UC San Diego Extended Studies students must follow the UC San Diego Extended Studies Academic Integrity Policy, which oversees all instances of academic misconduct, including but not limited to plagiarizing, cheating on exams, allowing someone access to your online course, improper or missing citations in coursework, and using artificial intelligence (see below).

About using artificial intelligence (AI) in Extended Studies courses

The Division of Extended Studies values the individual effort and authentic work done by students themselves. Therefore, unless otherwise stated, **using artificial intelligence to complete any portion of your coursework, whether it is an assignment, discussion, or quiz, is prohibited.** Unauthorized use of AI tools is a violation of the Extended Studies Academic Integrity Policy, Section IV, F, under which AI tools are classified as prohibited study aids. Please read Can I use artificial intelligence tools to complete my coursework? for more information.

Weighted Grading Criteria

Your grade in this course will be weighted according to the following criteria.

Discussion Board participation:	10%
Lab Assignments	40%
Quizzes	30%
Final project.	20%
TOTAL	100%

Click on 'Help' in the universal navigation menu anywhere in Canvas to learn about checking your grades.

Weighted Grading Criteria Explained

Read the descriptions about the types of graded work you will complete in this course.

Discussion Board: Various kinds of topics and questions will be discussed in the discussion board. I encourage you to ask questions and participate in answering other students' questions.

- Assignments: There will be total of 6 lab assignments. I expect students to use tools such as "Zip" to archive the project and submit.
- Quizzes: There will be 7 quizzes. It will include short answers and multiple-choice questions.
- Final: The details of the final project will be shared in the final module

Late Work Policy

Let me know ahead of time if you will be late in submitting your assignments, so that proper arrangements for grading can be made.

UC San Diego Extended Studies Policies and Resources

MyExtension

Your MyExtension account is your student records portal. Log into MyExtension to enroll in a course, drop a course, request verification of enrollment, request official transcripts and more.

Academic Policies and Procedures

Please refer to the UC San Diego Extended Studies Website (Student Resources tab) for specific details about academic policies and procedures. Navigate to the Grades section for grade information.

Conduct Code

All UC San Diego Extended Studies students are part of the UC San Diego community and are expected to follow University and UC-wide policies, including the Standards of Conduct and the UC San Diego Principles of Community. Reports of alleged violations involving sex offenses, including sexual assault and sexual misconduct, will be handled under the policies and procedures set forth in the University of California's Sexual Violence and Sexual Harassment Policy.

Emergencies on Campus

In the event of an emergency, information will be posted on the UC San Diego Extended Studies Website. Extended Studies students must access the website to find out the status of the emergency situation. Email and or phone lines may not be accessible. Information will be updated online as the situation progresses and an ALL CLEAR will be posted on the website once the situation is resolved.

Services for Students with Disabilities

UC San Diego Extended Studies is committed to providing equal access and an exceptional learning environment for all students. If you have any problems accessing course material, or if you anticipate or experience physical or academic barriers based on disability, we encourage you to contact our Services for Students with Disabilities Office to apply for reasonable accommodations. You can reach this office by email at unex-ssd@ucsd.edu or by calling (858) 822-1366.